

Monday Warm-Up: Unit 5, Work and Energy

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1 Memory Bank

- Let $\vec{u} = u_x\hat{i} + u_y\hat{j}$, and $\vec{v} = v_x\hat{i} + v_y\hat{j}$, and θ be the angle between them. The *dot-product* of $\vec{u}$ and $\vec{v}$ is

$$\vec{u} \cdot \vec{v} = u_x v_x + u_y v_y = |\vec{u}| |\vec{v}| \cos \theta \quad (1)$$

- $\hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{i} \cdot \hat{j} = 0$.
- Definition of the work, W , done by a net external force $\vec{F}$ through a displacement $\Delta\vec{x}$ is:

$$W = \vec{F} \cdot \Delta\vec{x} = F \Delta x \cos \theta \quad (2)$$

- Note the units are the product of force and displacement, so Newton meters. One Newton meter is called a Joule.
- When work W is done on a system, it gains *kinetic energy* KE :

$$W = \Delta KE = KE_f - KE_i \quad (3)$$

$$KE = \frac{1}{2}mv^2 \quad (4)$$

2 Work and Energy

- Evaluate the dot product, $\vec{u} \cdot \vec{v}$:

- A: $\vec{u} = 4\hat{i}, \vec{v} = 5\hat{j}$.
- B: $\vec{u} = 4\hat{i}, \vec{v} = 4\hat{i}$.
- C: $\vec{u} = 4\hat{i} + 2\hat{j}, \vec{v} = -2\hat{i} + 4\hat{j}$.

- Suppose a person pushes a box with a force $\vec{F} = 40\hat{i} - 30\hat{j}$ N of force, and the box moves $\Delta\vec{x} = 5\hat{i}$ m. (a) What is the work done? (b) Draw a diagram of the crate, force vector, and displacement vector. (c) What is the angle between $\vec{F}$ and $\Delta\vec{x}$? (d) What can the person do to increase the work done?

- A shopper pushes a grocery cart 20.0 m at constant speed on level ground, against a 35.0 N frictional force. He pushes in a direction 25.0° below the horizontal. (a) What is the work done on the cart by friction? (b) What is the work done on the cart by the gravitational force? (c) What is the work done on the cart by the shopper? (d) Find the force the shopper exerts, using energy considerations. (e) What is the total work done on the cart?

- Suppose a baseball player hits a baseball (0.15 kg) dropped in front of them with a baseball bat. (a) If the force during the hit is 1000 N, and the baseball is displaced 10 cm during the hit, what is the work done? (b) What is the change in kinetic energy, if we assume the baseball was initially stationary? (c) What is the final velocity of the ball? (d) If the ball leaves at a 45 degree angle, what is the range?